

Major Cryptocurrencies Overview

BTC, ETH, ISO 20022, DeFi, and DePIN

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Algorithm Basics and BW

What We'll Cover Today

1.  Bitcoin (BTC) - Digital gold
2.  Ethereum (ETH) - Smart contract platform
3.  ISO 20022 tokens (XRP, XLM, ALGO)
4.  DeFi tokens (UNI, AAVE)
5.  DePIN tokens (FIL, HNT)
6.  Hands-on: Blockchain explorer tour

Quick Recap: Cryptocurrency Basics

Last class we learned:

- ▶  Cryptocurrency = Digital money on blockchain
- ▶  Private keys control your crypto
- ▶  Benefits: Fast, global | Drawbacks: Volatile, complex
- ▶  Different wallet types for security

Today: Compare major cryptocurrencies in detail

Understanding Market Capitalization

What is market cap?

- ▶  Total value of all coins in circulation
- ▶ **Formula:** Price per coin $\times$ Number of coins
- ▶  Shows size and importance of cryptocurrency
- ▶ Helps compare different projects

Example:

- ▶ Bitcoin: \$50,000/coin $\times$ 19M coins = \$950B market cap
- ▶ Think of it like company valuation in stock market

Crypto Market Cap Visualization

Market cap rankings:

1.  Bitcoin (BTC)
2.  Ethereum (ETH)
3. Tether (USDT)
4. Ripple (XRP)
5. Solana (SOL)

Why it matters:

- ▶ Higher cap = More established
- ▶ Lower cap = Higher risk/reward



Bitcoin (BTC): The Original Cryptocurrency

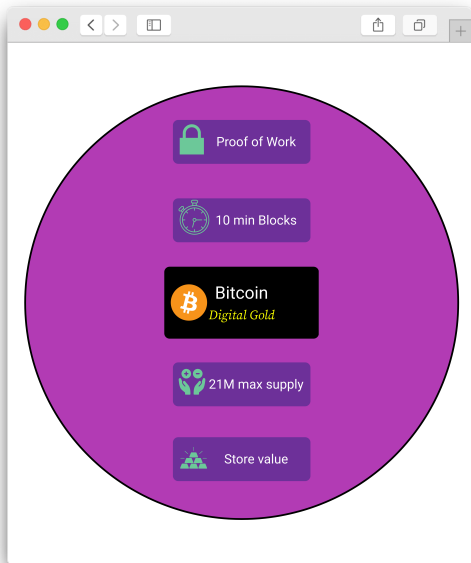
Key facts:

- ▶  Created in 2009 by Satoshi Nakamoto
- ▶  First cryptocurrency ever
- ▶  Maximum supply: 21 million coins
- ▶  Largest market cap (\$800B-\$1.5T)
- ▶  New block every 10 minutes

Purpose:

- ▶  Digital gold - store of value
- ▶  Peer-to-peer electronic cash

Bitcoin Features Deep Dive



What makes Bitcoin special:

- ▶  Most secure blockchain
- ▶  Most decentralized
- ▶  Accepted worldwide
- ▶  Fixed supply (deflationary)

Drawbacks:

- ▶  Slow (3-7 transactions/sec)
- ▶  High energy consumption
- ▶  Can be expensive to transfer

Bitcoin Use Cases

1.  **Store of value:** Like digital gold, hedge against inflation
2.  **International transfers:** Borderless payments
3.  **Financial sovereignty:** Control your own wealth
4.  **Investment:** Long-term appreciation potential
5.  **Banking alternative:** For unbanked populations

Real-world example:

- ▶ El Salvador adopted Bitcoin as legal tender (2021)
- ▶ Many companies hold Bitcoin on balance sheet (Tesla, MicroStrategy)

Ethereum (ETH): The World Computer

Key facts:

- ▶  Launched in 2015 by Vitalik Buterin
- ▶  First blockchain with smart contracts
- ▶  No maximum supply (inflationary with burning)
- ▶  Second largest market cap (\$300B-\$500B)
- ▶  New block every 12 seconds

Purpose:

- ▶  Platform for decentralized applications (dApps)
- ▶  Programmable blockchain

Ethereum vs Bitcoin

Bitcoin:

- ▶  Currency focus
- ▶  Simple transactions
- ▶  Store of value
- ▶  Slower, more secure
- ▶  Fixed supply

Ethereum:

- ▶  Platform focus
- ▶  Smart contracts
- ▶  Build applications
- ▶  Faster, more flexible
- ▶  No supply limit

Different purposes, both important!

Ethereum Smart Contracts in Action

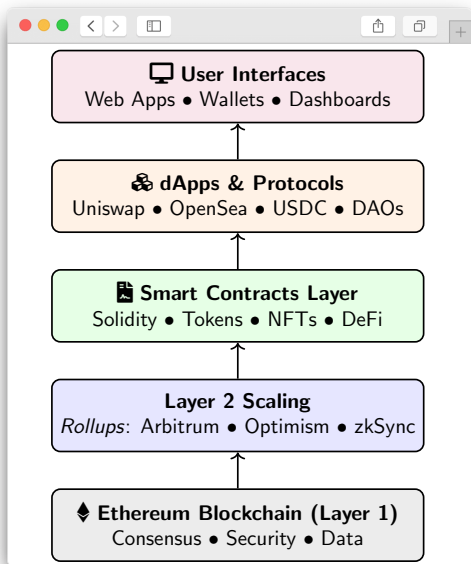
What can you build on Ethereum?

1.  **Tokens:** Create new cryptocurrencies (ERC-20)
2.  **NFTs:** Digital art and collectibles
3.  **DeFi apps:** Lending, borrowing, trading
4.  **Games:** Blockchain-based gaming
5.  **Identity:** Decentralized identity systems

Analogy:

- ▶ Bitcoin = Calculator (does one thing well)
- ▶ Ethereum = Smartphone (runs many apps)

Ethereum Ecosystem



Built on Ethereum:

- ▶  Uniswap (DEX)
- ▶  USDC (stablecoin)
- ▶  OpenSea (NFT marketplace)
- ▶  Thousands of tokens

Why it matters:

- ▶ Most developer activity
- ▶ Largest DeFi ecosystem
- ▶ Network effects

ISO 20022 Tokens: Banking Standard Compliance

What is ISO 20022?

- ▶  International standard for financial messaging
- ▶  Used by banks, payment systems, central banks
- ▶  Traditional finance adopting by 2025
- ▶  Some cryptos designed to be compliant

Why it matters for crypto:

- ▶  Bridge between crypto and traditional banking
- ▶  Regulatory compliance
- ▶  Cross-border payment infrastructure

Major ISO 20022 Compliant Tokens

1. **XRP (Ripple)**: Bank payment settlement, very fast
2. **XLM (Stellar)**: Cross-border payments, remittances
3. **ALGO (Algorand)**: Fast finality, carbon negative
4. **HBAR (Hedera)**: Enterprise-grade, low fees
5. **XDC (XinFin)**: Trade finance, supply chain

Common features:

- ▶  Very fast transaction speeds
- ▶  Low transaction fees
- ▶  Designed for financial institutions

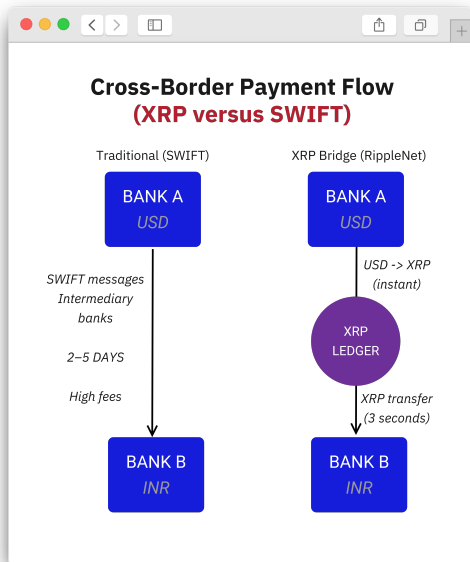
XRP (Ripple) Deep Dive

Key features:

- ▶ 🕒 3-5 second settlement
- ▶ 💰 \$0.0002 avg fee
- ▶ 🏦 Used by banks worldwide
- ▶ ↔ Bridge currency
- ▶ 📦 100B total supply (pre-mined)

Use case:

- ▶ Replace SWIFT for bank transfers
- ▶ Liquidity bridge between currencies



DeFi Tokens: Decentralized Finance

What is DeFi?

- ▶ 🏠 Banking services without banks
- ▶ </> Built with smart contracts
- ▶ 👤 Peer-to-peer financial services
- ▶ 🔒 You control your funds (non-custodial)

DeFi services:

1. 💵 Lending and borrowing
2. ↔ Decentralized exchanges (DEX)
3. 📊 Staking and yield farming
4. 🛡 Insurance protocols

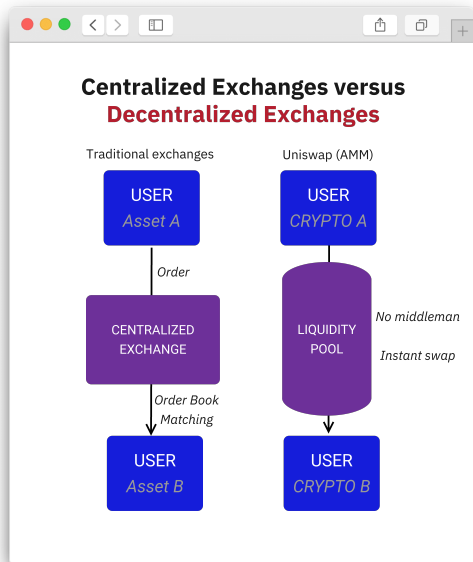
Major DeFi Tokens

1. **UNI (Uniswap)**: Largest decentralized exchange
2. **AAVE**: Lending and borrowing protocol
3. **MKR (MakerDAO)**: Creates DAI stablecoin
4. **LINK (Chainlink)**: Oracle network (real-world data)
5. **CRV (Curve)**: Stablecoin exchange

How they work:

- ▶  Governance tokens (vote on protocol changes)
- ▶  Earn fees from protocol usage
- ▶  Incentivize liquidity providers

Uniswap (UNI) Example



What is Uniswap:

- ▶ Trade tokens without exchange
- ▶ Automated market maker (AMM)
- ▶ Peer-to-peer trading
- ▶ You keep your keys

UNI token:

- ▶ Vote on protocol upgrades
- ▶ Share in trading fees

DePIN Tokens: Decentralized Physical Infrastructure

What is DePIN?

- ▶  Decentralized Physical Infrastructure Networks
- ▶  Use blockchain to coordinate real-world infrastructure
- ▶  Community-owned networks
- ▶  Token rewards for providing resources

Types of infrastructure:

1.  Wireless networks
2.  Storage networks
3.  Video streaming
4.  Mapping and location

Major DePIN Tokens

1. **ICP (Internet Computer):** Decentralized cloud computing
2. **XYO (XYO Network):** Decentralized location network
3. **RNDR (Render):** Distributed GPU rendering
4. **AR (Arweave):** Permanent data storage
5. **THETA:** Decentralized video streaming

How they work:

- ▶  Provide infrastructure (storage, bandwidth, compute)
- ▶  Earn tokens as rewards
- ▶  Users pay to use the network

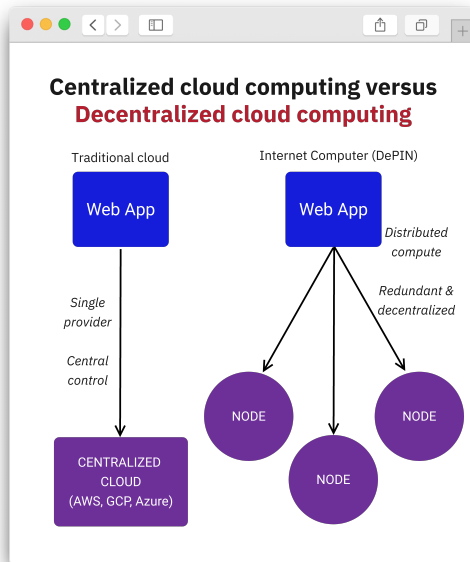
Internet Computer (ICP) Deep Dive

What is Internet Computer:

- ▶  Decentralized cloud computing
- ▶  Alternative to AWS, Google Cloud, Azure
- ▶  Anyone can provide computing
- ▶  Earn ICP for providing compute

Why it matters:

- ▶  Censorship resistant
- ▶  Cheaper
- ▶  Fully decentralized



Comparing All Categories

	BTC	ETH	ISO 20022	DeFi	DePIN
Purpose	Currency	Platform	Banking	Finance	Infrastructure
Speed	Slow (10m)	Medium (12s)	Very fast (3s)	Varies	Varies
Use case	Store value	Build apps	Payments	Banking alt	Physical network
Example	BTC	ETH	XRP, XLM	UNI, AAVE	ICP, XYO
Complexity	Simple	Complex	Medium	Complex	Complex

Each serves a different purpose in the crypto ecosystem

Hands-On Activity: Blockchain Explorer Tour

We'll explore multiple blockchains:

1.  Bitcoin: <https://blockchain.com>
2.  Ethereum: <https://etherscan.io>
3. XRP: <https://xrpscan.com>
4. Internet Computer: <https://dashboard.internetcomputer.org>

What to look for:

- ▶  Recent transactions
- ▶  Block time differences
- ▶  Transaction fees
- ▶  Network activity

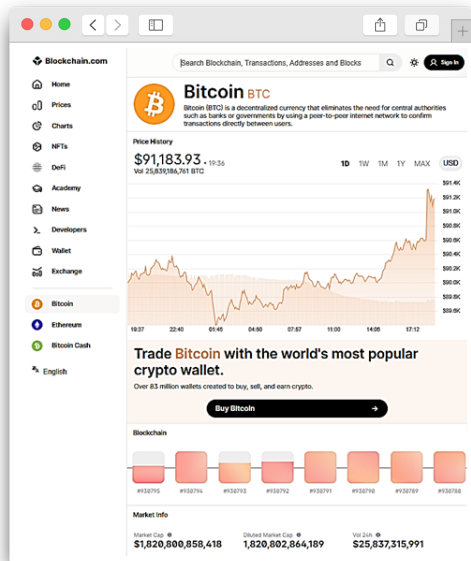
Exploring Bitcoin Blockchain

On blockchain.com:

1. 🖥️ Open blockchain.com
2. 🖱️ Click latest block
3. 👁️ View transactions
4. 💰 Check transaction fees
5. 👥 See how many confirmations

Notice:

- ▶ Block time 10 minutes
- ▶ Limited transactions per block



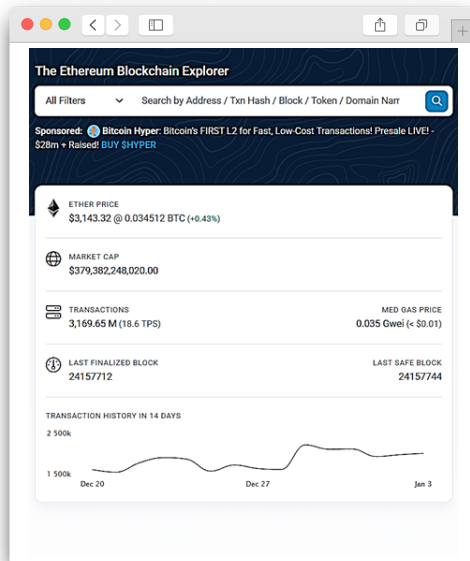
Exploring Ethereum Blockchain

On etherscan.io:

1.  Open etherscan.io
2.  View latest block
3.  See smart contract calls
4.  Check gas fees
5.  Browse token transfers

Notice:

- ▶ Block time 12 seconds
- ▶ Many more transactions
- ▶ Smart contract interactions



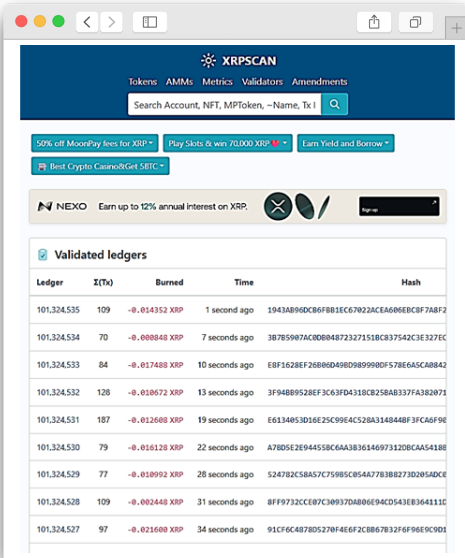
Exploring XRP Blockchain

On xrpscan.com:

1.  Open xrpscan.com
2.  View latest ledger
3.  Notice 3-5 second close time
4.  Check minimal fees
5.  See payment volume

Notice:

- ▶ Much faster than BTC/ETH
- ▶ Very low fees
- ▶ Focus on payments



The screenshot shows the XRPScan website interface. At the top, there's a navigation bar with links for Tokens, AMMs, Metrics, Validators, and Amendments. Below this is a search bar labeled "Search Account, NFT, MPToken, ~Name, Tx ID". There are several promotional banners for MoonPay, Slots, and NEXO. The main section is titled "Validated ledgers" and contains a table with the following data:

Ledger	I(Tx)	Burned	Time	Hash
101,324,535	109	-0.014352 XRP	1 second ago	1943AB96DCB6F8B1EC67022ACEA606EBC8F7ABF2
101,324,534	70	-0.000848 XRP	7 seconds ago	3B7B5907AC0B048723271518C37542C3E327EC
101,324,533	84	-0.017488 XRP	10 seconds ago	E8F1628EF26B06D49BD98990DF578E6A5CA8A2
101,324,532	128	-0.010672 XRP	13 seconds ago	3F94B89528EF3C63FD4318CB258AB337FA382071
101,324,531	187	-0.012608 XRP	19 seconds ago	E6134053D16E25C99E4C528A314844BF3CA6F9E
101,324,530	79	-0.016128 XRP	22 seconds ago	A7BD5E2E944558C6AA3B36146973120BCAA5418B
101,324,529	77	-0.010992 XRP	28 seconds ago	524782C58A57C759B5C054A77B3B273D205ADCE
101,324,528	109	-0.002448 XRP	31 seconds ago	8FF9732CCE07C30937DAB06E94C0543E8364111C
101,324,527	97	-0.021600 XRP	34 seconds ago	91CF6C487BD5270F4E6F2CBB67B32F6F96E9C9D1

Comparing What You See

Key differences observed:

- ▶ 🕒 **Speed:** BTC (10m) vs ETH (12s) vs XRP (3s)
- ▶ 🏠 **Fees:** XRP lowest, ETH variable, BTC moderate
- ▶ ↔ **Transactions:** ETH handles most types
- ▶ </> **Complexity:** BTC simple, ETH complex (smart contracts)
- ▶ 👤 **Activity:** All show live network usage

Key insight:

- ▶ Different blockchains optimized for different purposes
- ▶ Trade-offs between speed, security, decentralization

Understanding Transaction Details

What you can see in any transaction:

1. # Transaction hash (unique ID)
2. 👤 From address (sender)
3. 👤+ To address (receiver)
4. 💰 Amount transferred
5. 🏠 Fee paid
6. 🕒 Timestamp
7. ✅ Confirmations

Try this: Click on any transaction and explore!

Portfolio Diversification Concept

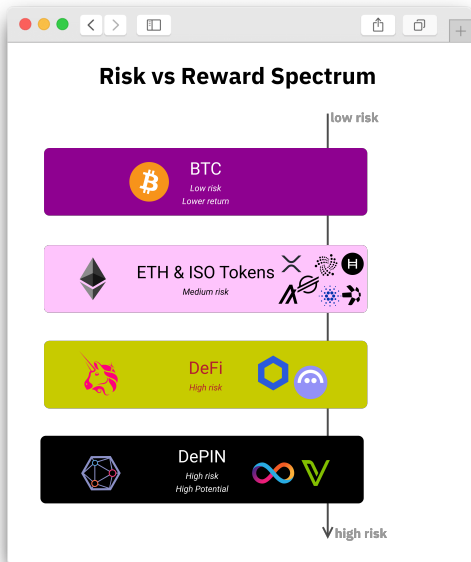
Don't put all eggs in one basket:

- ▶  BTC: 40-50% (stable, established)
- ▶  ETH: 20-30% (growth + utility)
- ▶ ISO 20022: 10-15% (banking integration)
- ▶ DeFi: 10-15% (innovation, higher risk)
- ▶ DePIN: 5-10% (emerging sector)

Example allocation:

- ▶ This is NOT financial advice!
- ▶ Just showing diversification concept
- ▶ Higher risk = potentially higher reward

Risk vs Reward Spectrum



General principle:

- ▶  Lower risk = More established
- ▶  Higher risk = More potential upside
- ▶  All crypto is volatile

Best practices:

- ▶ Start with BTC/ETH
- ▶ Research before investing
- ▶ Never invest more than you can lose

Key Takeaways

1.  Bitcoin = Digital gold, store of value, most secure
2.  Ethereum = Platform for dApps, smart contracts
3. ISO 20022 = Banking compliant, fast payments (XRP, XLM)
4. DeFi = Banking without banks (UNI, AAVE)
5. DePIN = Decentralized infrastructure (ICP, XYO)

Remember:

- ▶ Each category serves different purpose
- ▶ Explore blockchain explorers to see activity
- ▶ Diversification reduces risk

Questions and Discussion